

Exercise #1

$S=\{a,b,c,d,e,f,g,h,i,j\}$

$A(a)=\{1,2,3\}$

$A(b)=\{1,2,3\}$

$A(c)=\{1,2,3\}$

$A(d)=\{1,2,3\}$

$A(e)=\{1,2\}$

$A(f)=\{1,2\}$

$A(g)=\{1,2\}$

$A(h)=\{1\}$

$A(i)=\{1\}$

$A(j)=\{\}$

T:SxA→S

$T(a,1)=b$

$T(a,2)=c$

$T(a,3)=d$

$T(b,1)=e$

$T(b,2)=f$

$T(b,3)=g$

$T(c,1)=e$

$T(c,2)=f$

$T(c,3)=g$

$T(d,1)=e$

$T(d,2)=f$

$T(d,3)=g$

$T(e,1)=h$

$T(e,2)=i$

$T(f,1)=h$

$T(f,2)=i$

$T(g,1)=h$

$T(g,2)=i$

$T(h,1)=j$

$T(i,1)=j$

$R(a,1)=-2$

$R(a,2)=-4$

$R(a,3)=-2$

$R(b,1)=-7$

$R(b,2)=-4$

$R(b,3)=-6$

$R(c,1)=-3$

$R(c,2)=-2$

$R(c,3)=-4$

$R(d,1)=-4$

$R(d,2)=-1$

$R(d,3)=-5$

$R(e,1)=-1$

$$R(e,2)=-4$$

$$R(f,1)=-6$$

$$R(f,2)=-3$$

$$R(g,1)=-3$$

$$R(g,2)=-5$$

$$R(h,1)=-3$$

$$R(i,1)=-4$$

Exercise #2

$$S = \{(y, x) \mid y=0,1,2; x=0,1,2\}$$

$$A = \{(-1,0), (1,0), (0,-1), (0,1)\}$$

Actions are up, down, left, right

$$T(s, a) = \begin{cases} s+a & \text{if } s+a \text{ in } [0;2]^2 \\ s & \text{otherwise} \end{cases}$$

$$R(s, a) = \begin{cases} 5 & \text{if } T(s, a) \text{ in } \{(0,0), (1,2), (2,1)\} \\ -1 & \text{otherwise} \end{cases}$$

Exercise #3

Q1:

$S=\{a,b,c,d,e,f\}$

$A(a)=\{d\}$

$A(b)=\{u,d\}$

$A(c)=\{l\}$

$A(d)=\{u,l,r\}$

$A(e)=\{u,d,r\}$

$A(f)=0$

$T(a,d)=e$

$T(b,u)=f$

$T(b,d)=d$

$T(c,l)=d$

$T(d,u)=b$

$T(d,l)=e$

$T(d,r)=c$

$T(e,d)=f$

$T(e,u)=a$

$R(s,a) = \begin{cases} 100 & \text{if } T(s,a)=f \\ 0 & \text{otherwise} \end{cases}$

Q2:

We assume:

$\gamma=0,5 \quad \alpha=0,5$

F is terminal

Prior to learning,

Q	a	b	c	d	e	f
u	0	0	0	0	0	0
d	0	0	0	0	0	0
l	0	0	0	0	0	0
r	0	0	0	0	0	0

Episode 1

Step01 $s_0=e$ $a_0=d$ $Q(e,d)=0+0,5*(100+0,5*0-0)=50$
 $s_1=f$
End

Episode 2

Step01 $s_0=b$ $a_0=u$ $Q(b,u)=0+0,5*(100+0,5*0-0)=50$
 $s_1=f$
End

Episode 3

Step01	s0=d	a0=u	$Q(d,u)=0+0,5*(0+0,5*\max\{0,50\}-0)=12,5$
Step02	s1=b	a1=u	$Q(b,u)=50+0,5*(100+0,5*0-50)=75$
	s2=f		
	End		

So after 3 episodes:

Q	a	b	c	d	e	f
u	0	75	0	12,5	0	0
d	0	0	0	0	50	0
l	0	0	0	0	0	0
r	0	0	0	0	0	0